ECON 3640–001: PROBABILITY & STATISTICAL INFERENCE

Fall 2024

Instructor: Ayixunbati Nuerlantai (Eisen)
Office Hours: By appointment
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Time: MW 1:25-2:45pm
Class Room: GC 2675

Overview

In this course, you will be exposed to statistical analysis from the lens of an economist. You will learn graphical and numerical methods of summarizing data, the concept of probability, and hypothesis testing. You will find these skills useful in a wide range of contexts ranging from reading and interpreting news articles critically, becoming an educated consumer, evaluating policies, and preparing you for advanced courses in quantitative analysis. This course belongs to the category of Quantitative Reasoning (QR-B) courses and is a pre-requisite to take ECON 4650.

Required Materials

Layth C. Alwan; Bruce A. Craig; George P. McCabe, *The Practice of Statistics for Business and Economics*, 5th edition, Macmillan learning, 2020.

You should have access to the digital version of this book through the *Bookshelf* tab in Canvas. The older edition of this book is also acceptable, but be aware that the order of the chapters and some content may be a bit different.

The textbook can be found at:

- Inclusive Access Program (*Bookshelf* in Canvas)
- Campus Store/ University Library
- www.used.addall.com

Software Requirements

Our classes will always have *theoretical* and *applied* components. The latter will be done using R, an open-source and completely *free* statistical/programming language. In applied lectures, I use the RStudio IDE (Integrated Development Environment), also freely available. However, students are welcome to use the platform they prefer.

In addition, having a spreadsheet editor (e.g., Excel) is strongly recommended. University of Utah students should have free access to Microsoft Office programs through `Office 365`.

No previous knowlegde in R is required. The course will guide students through every step in the applied lectures.

- Download R: `https://cloud.r-project.org/`
- Download RStudio: `https://rstudio.com/products/rstudio/`

Prerequisites

College Algebra, MATH 1090, ECON 2010 and 2020.

Course Outcomes

At the completion of this course, you will be able to:

1. Comfortably use R to perform basic statistical analyses;
2. Critically assess the most common tools in Statistics;
3. Understand the difference between causation and correlation in empirical data and policy debates;
4. Interpret and critique the empirical results in Economics journals and newspaper articles.

Assignments, Exams, and Grading

- **Assignments** (30%): Whenever enough content is seen in class, a new problem set will be uploaded to our Canvas course page. There will be a set of six assignments (5% each). In order to get full credit, students are required to turn the assignment in time. Please check more information in the *Late Policy* section, on the next page.
- **Midterm exam** (30%): The midterm exam will ask theoretical and applied questions on descriptive statistics and some probability concepts.
- **Final exam** (40%): The Final exam is comprehensive, covering all studied topics. In principle, the final exam will also be take-home. We will discuss an appropriate window allowing every student to have time to complete the exams.

Important note 1: Exams will strongly reflect what you will practice in class and in Problem Sets.

Important note 2: For *any* assignment, if an answer contains a direct "Copy+Paste" from the lecture notes, or if it is copied from someone else's work, **no credit will be given**. Read more in the *Academic Dishonesty* section, on page 5.

Late Policy:

You are expected to turn things in/on time. It is your responsibility to maintain your computer and related equipment in order to participate in this course. Equipment failures will not be an acceptable excuse for late or absent assignments. Similarly, **it is your responsibility to start assignments early enough** so that even if you are in traffic, your flight gets delayed, you are called into work, you do work for another class, etc., you still have time to deal with the situation and then finish the assignment.

The late assignments will be accepted up to 24 hrs after the due date with a **20% late penalty**. Any late submissions later than 24 hrs will receive no credit.

Important Dates

- **Classes begin**: Aug 19
- **Last day to add/drop classes**: Aug 30
- **Labor Day Holiday**: Sep 2
- **Midterm Exam**: Oct 3-4 (tentative)
- **Fall Break**: Oct 6–13
- **Last day to withdraw from classes**: Oct 18
- **Thanksgiving Break**: Nov 28-Dec 1
- **Classes End**: Dec 5
- **Final Exam**: Dec 9-10 (tentative)

Letter Grade Distribution

Performance	*Letter Grade*
Superior performance	A (93-100%), A- (90-92.9%)
Good performance	B+ (87-89.9%), B (83-86.9%), B- (80-82.9%)
Standard performance	C+ (77-79.9%), C (73-76.9%), C- (70-72.9%)
Substandard performance	D+ (67-69.9%), D (63-66.9%), D- (60-62.9%)
Unsatisfactory performance	E (0-59.9%)

Class Policies

- Students are strongly encouraged to attend the class, although attendance is not a part of their grades;
- The exams will reflect what has been studied in class and viewed in questiosn from Assignments;
- Please be respectful of others in the class. If you have to make a call or talk to someone for whatever reason, you are welcome to step out of the class for a moment;
- You can expect me to reply to emails/messages within 24 hours on weekdays and 48 hours on weekends/holidays. If I do not reply within this time, please do not hesitate to reach out again. In a very rare case, I may miss your email by accident;

- Foods and drinks are allowed in the class without disturbing others;
- Please notify me in the event of an emergency that prevents you from doing an exam or completing the course;
- No make-up exams will be given for reasons other than a documented medical emergency or conflict with another exam.

Tentative Course Schedule

Week	Content	Note
Week 1	Introduction	Syllabus + R
Week 2	Ch1:Examining Distributions	A1
Week 3	Ch2:Examining Relationships	Labor day (Sep 2)
Week 4	Ch2,Ch3: Producing data	A2
Week 5	Ch3	
Week 6	Ch4:Probability	A3
Week 7	Ch4	Midterm (Oct 3-4)
Week 8	Fall break	Fall break
Week 9	Ch5:RV&Probability Distribution	
Week 10	Ch5,Ch6:Sampling Distribution	A4
Week 11	Ch7:Introduction to Inference	
Week 12	Ch8:Inference - Means	A5
Week 13	Ch10:Inference - Proportions	
Week 14	Ch11:Inference - Categorical data	
Week 15	Ch17:Nonparametric Tests	Thanksgiving/A6
Week 16	Review	
Dec 9-10	Final exam	Good luck!

Note. A = Assignment

Important Note

This syllabus is meant to serve as an outline and guide for our course. Please note that I may modify it with reasonable notice to you. I may also modify the Course Schedule to accommodate the needs of our class. Any changes will be announced in class and posted on Canvas under *Announcements*.

Institutional Policies and Procedures

Faculty and Student Responsibilities

All students are expected to maintain professional behavior in the classroom setting, according to the Student Code, spelled out in the Student Handbook. Students have specific rights in the classroom as detailed in Article III of the Code. The Code also specifies proscribed conduct (Article XI) that involves cheating on tests, plagiarism, and/or collusion, as well as fraud, theft, etc. Students should read the Code carefully and know they are responsible for the content. According to Faculty Rules and Regulations, it is the faculty responsibility to enforce responsible classroom behaviors, beginning with verbal warnings and progressing to dismissal from class and failing grade. Students have the right to appeal such action to the Student Behavior Committee. "Faculty . . . must strive in the classroom to maintain a climate conducive to thinking and learn- ing." PPM 8-12.3, B.

"Students have a right to support and assistance from the University in maintaining a climate conducive to thinking and learning." PPM 8-10, II. A.

Wellness Statement

Personal concerns such as stress, anxiety, relationship difficulties, depression, cross-cultural differences, etc., can interfere with a students ability to succeed and thrive at the University of Utah. For helpful resources contact the Center for Student Wellness: www.wellness.utah.edu, 801-581-7776.

Student Names & Personal Pronouns

Class rosters are provided to the instructor with the students legal name as well as "Preferred first name" (if previously entered by you in the student profile section of your CIS account). While CIS refers to this as merely a preference, I will honor you by referring to you with the name and pronoun that feels best for you in class, on papers, exams, group projects, etc. Please advise me of any name or pronoun changes (and update CIS) so I can help create a learning environment in which you, your name, and your pronoun will be respected.

Academic Dishonesty

The instructor of this course will take appropriate actions in response to Academic Dishonesty, as defined the Universitys Student Code. Acts of academic dishonesty include but are not limited to:

- Cheating: using, attempting to use, or providing others with any unauthorized assistance in taking quizzes, tests, examinations, or in any other academic exercise or activity. Unauthorized assistance includes:
  - Working in a group when the instructor has designated that the quiz, test, examination, or any other academic exercise or activity be done individually;
  - Depending on the aid of sources beyond those authorized by the instructor in writing papers, preparing reports, solving problems, or carrying out other assignments;
  - Substituting for another student, or permitting another student to substitute for oneself, in taking an examination or preparing academic work;
  - Acquiring tests or other academic material belonging to a faculty member, staff member, or another student without express permission;
  - Continuing to write after time has been called on a quiz, test, examination, or any other academic exercise or activity;

  - Submitting substantially the same work for credit in more than one class, except with prior approval of the instructor; or engaging in any form of research fraud.
- Falsification: altering or fabricating any information or citation in an academic exercise or activity.
- Plagiarism: representing, by paraphrase or direct quotation, the published or unpublished work of another person as ones own in any academic exercise or activity without full and clear acknowledgment. It also includes using materials prepared by another person or by an agency engaged in the sale of term papers or other academic materials.

Sexual Harassment

Title IX of the Education Amendments of 1972 prohibits sex discrimination against any participant in an educational program or activity that receives federal funds. The act is intended to eliminate sex discrimination in education. Title IX covers discrimination in programs, admissions, activities, and student-to-student sexual harassment. It is the policy of the University of Utah to maintain an academic and work environment free of sexual harassment for students, faculty, staff and participants. A claim under this policy may be brought by any faculty, staff member, student or participant based on the conduct of any University employee or student that is related to or in the course of University business. A claim may also be brought by an administrator acting on behalf of the University. The Office of Equal Opportunity and Affirmative Action (OEO/AA) will handle all alleged sexual harassment matters pursuant to the Procedures set forth in Policy 5-210.

Any student, staff, faculty member, or participant in university services or activities who believes that there is or has been sexual harassment should contact any of the following:

- the cognizant academic chair of the department or the dean of the college within which the conduct occurred;
  - the immediate supervisor or director of the operational unit within which the conduct occurred;
  - the Human Resources Division (Address: 420 Wakara Way, Suite 105, Salt Lake City, Utah 84108; Phone: 801-581-2169; TDD: 801-585-9070);
  - directors or deans of Student Affairs and Services (Associate Dean - SW 108; Phone: 801-581- 8828);
  - or the Office of Equal Opportunity and Affirmative Action (201 South Presidents Circle, John Park Building, Room 135, Salt Lake City, Utah 84112; Phone/TDD: 801-581-8365).

Once informed, the supervisor, chair or dean shall consult with OEO/AA. Similarly, OEO/AA shall advise as appropriate the cognizant dean, staff director or administrator of any complaints of sexual harassment OEO/AA receives concerning a member of the faculty, staff or student body.

Withdrawal Policy and "I" Grade

Failure to withdraw from school results in a E or EU grade being recorded in all classes. Students may officially withdraw (*W*) from a class or all classes after the drop deadline through the midpoint of a course. A *W* grade is recorded on the transcript and appropriate tuition/fees are assessed. The grade of *W* is not used in calculating the students GPA.

An Incomplete grade can be given for work not completed due to circumstances beyond your control. You must be passing the course and have completed at least 80% of the required coursework. Arrangements must be made between you and the instructor concerning the completion of

the work. You may not retake a course without paying tuition. If you attend class during a subsequent term, in an effort to complete the coursework, you must register for the course. Once the work has been completed, the instructor submits the grade to the Registrars Office. The *I* grade will change to an *E* if a new grade is not reported within one year. A written agreement between you and the instructor may specify the grade to be given if the work is not completed within one year. Copies of the agreement are kept by the instructor and the academic department.

Americans with Disabilities Act (ADA) Statement

The University of Utah seeks to provide equal access to its programs, services and activities for people with disabilities. If you will need accommodations in the class, reasonable prior notice needs to be given to the Center for Disability Services, 162 Olpin Union Building, 581-5020 (V/TDD). CDS will work with you and the instructor to make arrangements for accommodations. All written information in this course can be made available in alternative format with prior notification to the Center for Disability Services.

Additionally, the University endeavors to provide reasonable accommodations and to ensure equal access to qualified persons with disabilities. Inquiries concerning perceived discrimination or requests for disability accommodations may be referred to the University's Title IX/ADA/Section 504 Coordinator: Director, Office of Equal Opportunity and Affirmative Action, 201 South Presidents Circle, Rm. 135, Salt Lake City, UT, 84112. 801-581-8365 (voice/tdd), 801-585-5746 (fax). www.oeo.utah.edu.